

6_tuples in Python mastering immutable data strucutre

Tuples are immutable and collection of ordered data elements by creating elements inside ().

```
mytuple=(1,2,3,4,5)
print(mytuple)
#Hold multiple data types in a tuple
mytuple1=(1,2.5,3+2j,"Hanu",[9,5.5,6],(1,2,3))
print(mytuple1)
```

[3] ✓ 0.0s

```
... (1, 2, 3, 4, 5)
     (1, 2.5, (3+2j), 'Hanu', [9, 5.5, 6], (1, 2, 3))
```

```
print(mytuple1[0])
print(mytuple1[3])
print(mytuple1[4][1])
```

[5] ✓ 0.0s

```
... 1
     Hanu
     5.5
```

```
print(mytuple1[4])
print(mytuple1[5])
print(type(mytuple1[4]))
print(type(mytuple1[5]))
print(type(mytuple1[3]))
```

[11] ✓ 0.0s

```
... [9, 5.5, 6]
     (1, 2, 3)
     <class 'list'>
     <class 'tuple'>
     <class 'str'>
```

▷ ▾

```
Mlist=[1,2,3,4,5]
Mtupple=(1,2,3,4,5)
Mlist[0]=100
print(Mlist)
Mtupple[0]=100
print(Mtupple)
```

[13] ⊗ 0.0s

```
... [100, 2, 3, 4, 5]
```

...

```
-----
TypeError                                Traceback (most recent call last)
/tmp/ipykernel_2432/4053957814.py in <cell line: 0>()
      3 Mlist[0]=100
      4 print(Mlist)
----> 5 Mtupple[0]=100
      6 print(Mtupple)
```

TypeError: 'tuple' object does not support item assignment

```
#unpacking a tuple
x, y, z, a, b = mytuple
print(x)
print(y)
print(z)
print(a)
print(b)
```

[14] ✓ 0.0s

```
... 1
    2
    3
    4
    5
```

```
new_tuple=(26,)
print(type(new_tuple))
```

[15] ✓ 0.0s

```
... <class 'tuple'>
```

```
new_tuple1=(10,20,10,10,20,30,40,30,50,60,70,80,90,100)
print(new_tuple1)
print(len(new_tuple1)) #prints the length of the tuple
print(new_tuple1.count(10)) #prints the number of occurrences of 10 in the tuple
print(new_tuple1.index(20)) #prints the index of the first occurrence of 20 in the tuple
print(new_tuple1.index(30)) #prints the index of the first occurrence of 30 in the tuple
print(new_tuple1.index(30, 4)) #prints the index of the first occurrence of 30 in the tuple starting from index 4
print(new_tuple1.index(30, 4, 10)) #prints the index of the first occurrence of 30 in the tuple starting from index 4 and ending at index 10
print(new_tuple1.index("Hanumanth")) #gives error because "Hanumanth" is not present in the tuple
```

[18] ✗ 0.0s

```
... (10, 20, 10, 10, 20, 30, 40, 30, 50, 60, 70, 80, 90, 100)
    14
    3
    1
    5
    5
    5
```

```
-----
ValueError                                Traceback (most recent call last)
/tmp/ipykernel_2432/832429188.py in <cell line: 0>()
      7 print(new_tuple1.index(30, 4)) #prints the index of the first occurrence of 30 in the tuple starting from index 4
      8 print(new_tuple1.index(30, 4, 10)) #prints the index of the first occurrence of 30 in the tuple starting from index 4 and ending at index 10
----> 9 print(new_tuple1.index("Hanumanth")) #gives error because "Hanumanth" is not present in the tuple

ValueError: tuple.index(x): x not in tuple
```

```
# Tuple concatenation
tuple1 = (1, 2, 3)
tuple2 = (4, 5, 6)
tuple3 = tuple1 + tuple2
print(tuple3) # Output: (1, 2, 3, 4, 5, 6)
```

[19] ✓ 0.0s

... (1, 2, 3, 4, 5, 6)



```
fruits = ("apple", "banana", "cherry", "date", "elderberry")
sub_fruits = fruits[1:4] # Slicing from index 1 to 3 (4 is not included)
print(sub_fruits) # Output: ('banana', 'cherry', 'date')
```

[20] ✓ 0.0s

... ('banana', 'cherry', 'date')

```
tuple1=('a','b','c','d','e','f','g')
print(tuple1)
print(tuple1[-3]) #prints the third last element of the tuple
print(tuple1[2:5]) #prints the elements from index 2 to 4
print(tuple1[2:]) #prints the elements from index 2 to the end
print(tuple1[:5]) #prints the elements from the start to index 4
```

[21]

... ('a', 'b', 'c', 'd', 'e', 'f', 'g')
e
('c', 'd', 'e')
('c', 'd', 'e', 'f', 'g')
('a', 'b', 'c', 'd', 'e')